

PyMutScan

Advanced DNA Mutation Detector

Select Reference Gene

BRCA1

NCBI Reference Sources

Gene: BRCA1 | ID: 672

RefSeq: NM_007294.4

Location: Chr 17q21.31

Full Length: 81,189 bp

Function: DNA damage repair, tumor suppression

[NCBI Gene Page](#)

[FASTA Sequence](#)

[ClinVar Variants](#)

[OMIM Entry](#)

Upload FASTA File

Upload .fasta or .txt

Drag and drop file here

Limit 200MB per file • FASTA, FA, TXT

Browse files

About

PyMutScan aligns patient DNA against a reference gene, detects SNPs, translates codons, designs primers, predicts drug resistance, visualizes mutation hotspots, and stores patient records in a local SQLite database.

PyMutScan — Advanced DNA Mutation Detector

Sequence alignment · Variant calling · Codon translation · Primer design · Drug resistance · Mutation heatmap · Patient records

Virtual Patient Manager — Create or Load a Patient Profile

Create New Patient

Load Existing Patient

Fill in patient details below. Gene and Sequence will be pulled from the current sidebar selection and query box when you click Save.

Patient Name *

navneet

Age

34

–

+

Gender

Male

Blood Group

A+

Medical History / Symptoms

Preliminary Diagnosis / Notes

e.g. Family history of breast cancer, BRCA1 mutation suspected

e.g. Referred for pharmacogenomics screening

The current Gene (sidebar) and Query Sequence (below) will be saved with this patient profile.

Save Patient Profile

Reference — BRCA1

ATGGATTATCTGCTCTTCGCGTTGAAGAAGTACAAAATGTC

67 bp demo segment · Source: NCBI GenBank NM_007294.4

> View Full Gene Info & Sources

Query Sequence (Patient)

Load Sample with Mutations

Paste DNA sequence or upload FASTA above (A, T, C, G only)

ATGGATTATCAGCTCTTCGCGTTGATGAAGTACAAAATGTCATGAATGCTATGCAGAAAATCTTAG

67 bp entered

Run Mutation Analysis

Analysis complete — 3 variant(s) detected

Total Variants

3

Sequence Identity

95.5%

Pathogenic

1

Uncertain

1

Benign

1

Alignment

Mutations Table

Sequence Stats & GC

Codon Translation

Mutation Heatmap

Primers

Detected Variants — BRCA1

localhost:8501

1/2

	Position	Codon#	Ref	Alt	Type	Eff
0	12	4	T	A	SNP	Mis
1	27	9	A	T	SNP	Sil
2	45	15	T	G	SNP	Mis

 Download Results as CSV

Patient Records & Database

Persistent SQLite storage — all patient profiles and analysis results are saved locally in `pymutscane_patients.db`

-  Patient Profiles
-  Analysis History
-  **Export Data**

Export All Data

-  Patient Profiles
-  Analysis History

No patient data to export.

No analysis data to export.

Combined Full Export (JSON)

Database is empty. No data to export yet.

 Database is stored locally as `pymutscane_patients.db` in the same directory as this script. Back it up regularly.